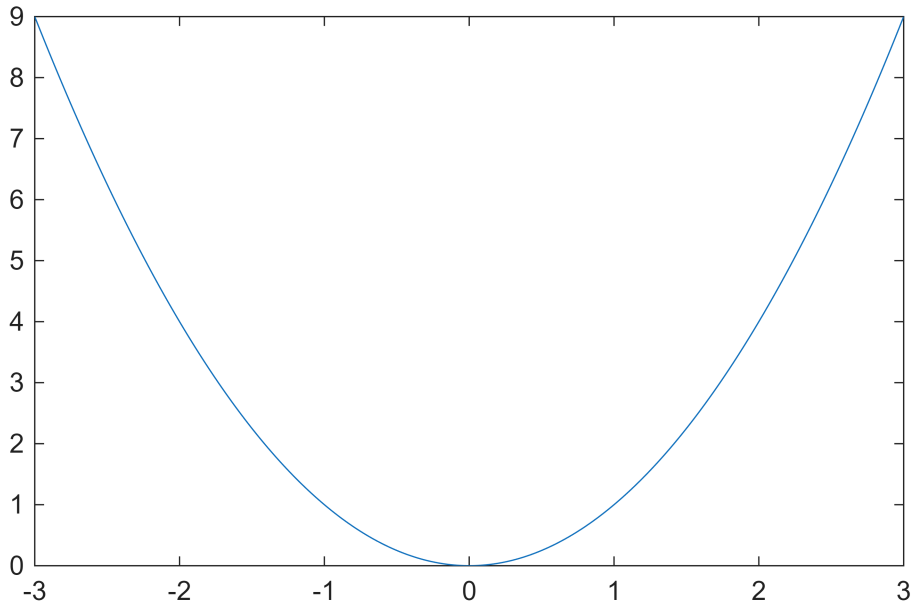
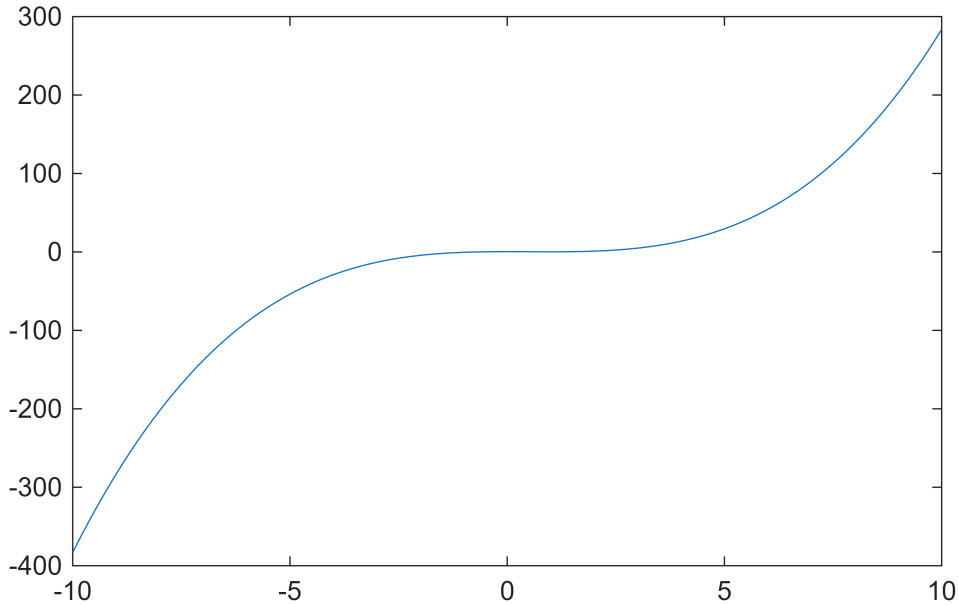
ques 1

```
x=-3:0.01:3;%define the range of x
y=x.^2;%compute y=x^2
plot(x,y);%plot the function
```



ques 2

```
x=-10:0.01:10;%define the range of x
y=(x.^3)/3-(x.^2)/2+1/3;%compute y=x^3/3 -x^2/2+1/3
plot(x,y);%plot the function
```



ques 3

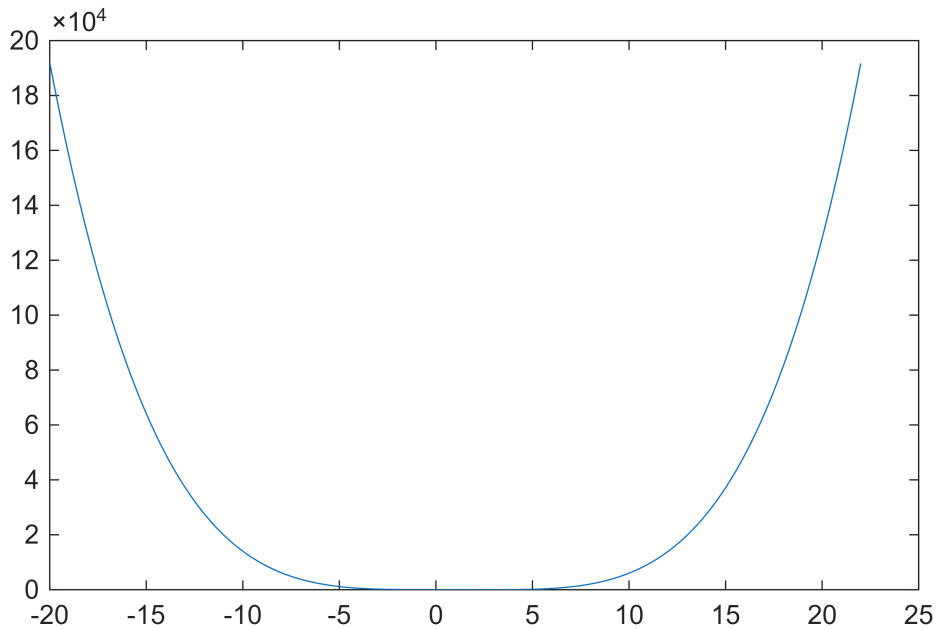
```
x=-20:0.01:22
```

```
x = 1×4201
  -20.0000  -19.9900  -19.9800  -19.9700  -19.9600  -19.9500  -19.9400  -19.9300 ···
```

```
y=(x.^4)-4*(x.^3)+10
```

```
y = 1×4201
10^5 ×
    1.9201    1.9164    1.9128    1.9091    1.9054    1.9018    1.8981    1.8945 ···
```

```
plot(x,y)
```



ques4

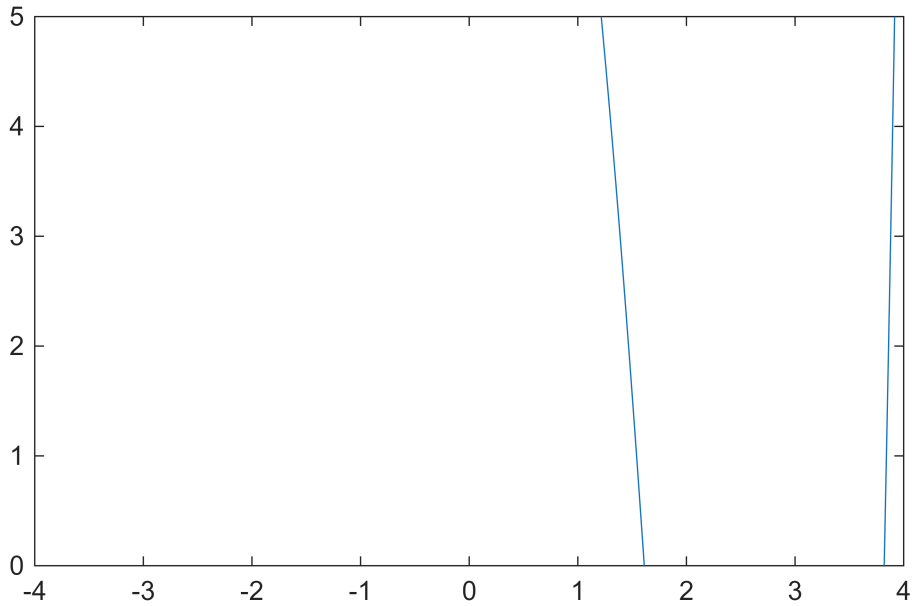
```
x=-4:0.01:4
```

```
x = 1×801
   -4.0000   -3.9900   -3.9800   -3.9700   -3.9600   -3.9500   -3.9400   -3.9300 ···
```

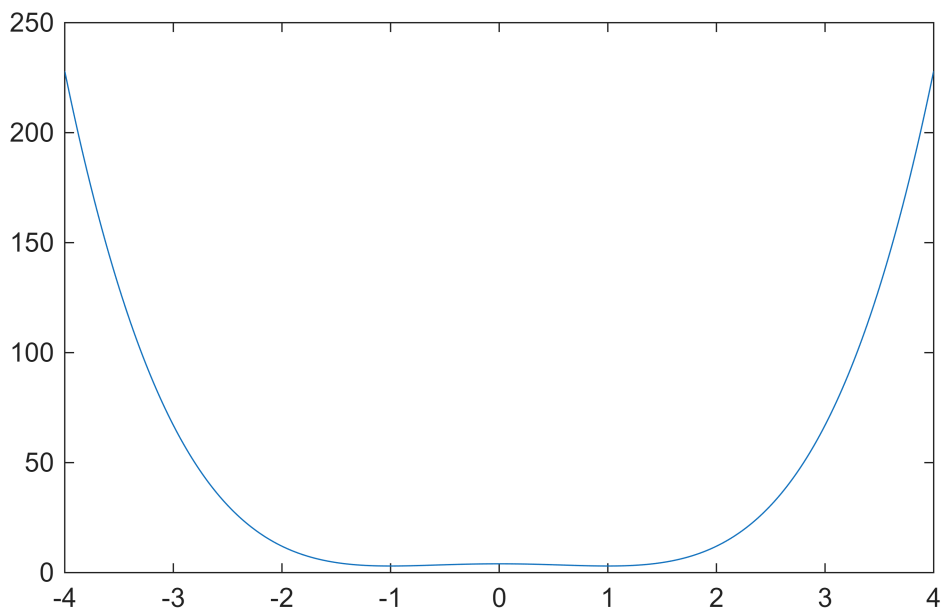
```
y=(x.^4)-2.*(x.^2)+4
```

```
y = 1×801
  228.0000  225.6094  223.2375  220.8842  218.5494  216.2330  213.9350  211.6551 ···
```

```
xlim([-4 4])%sets x axis limits
ylim([0 5])%sets y axis limits
xticks(-4:1:4)%sets x axis ticks with an increment of 1
yticks(0:1:5)%sets y axis ticks with an increment of 1
```




```
plot(x,y)
```



ques 5

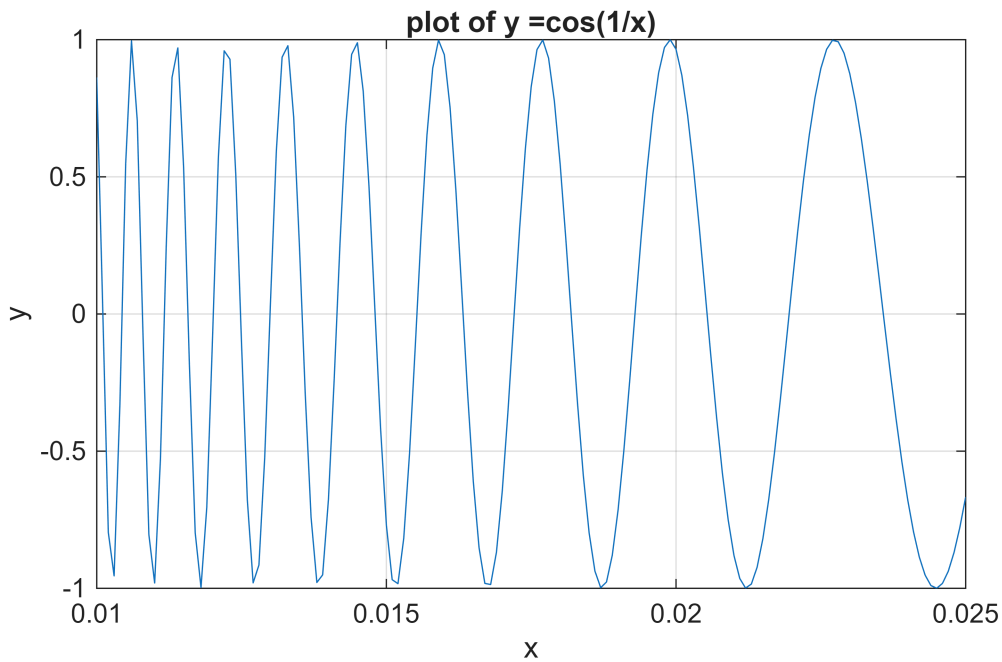
```
x=1/100:0.0001:1/40
```

```
x = 1×151
    0.0100    0.0101    0.0102    0.0103    0.0104    0.0105    0.0106    0.0107 ···
```

```
y =cos(1./x)
```

```
y = 1×151
    0.8623    0.0497   -0.7962   -0.9547   -0.3290    0.5484    0.9958    0.7040 ···
```

```
plot(x,y)
xlabel('x')
ylabel('y')
title('plot of y =cos(1/x)')
grid on %displays grid
```



ques 6

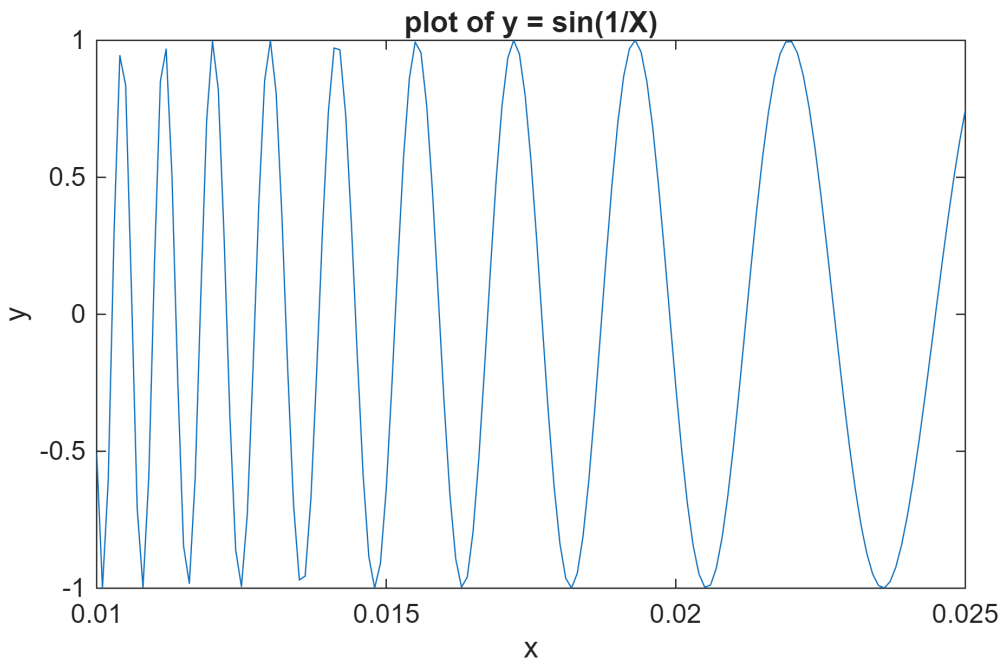
```
x=1/100:0.0001:1/40
```

```
x = 1×151
    0.0100    0.0101    0.0102    0.0103    0.0104    0.0105    0.0106    0.0107 ···
```

```
y=sin(1./x)
```

```
y = 1×151
   -0.5064   -0.9988   -0.6051    0.2974    0.9443    0.8362    0.0917   -0.7102 ···
```

```
plot(x,y)
xlabel('x')
ylabel('y')
title('plot of y = sin(1/X)')
```



ques 7

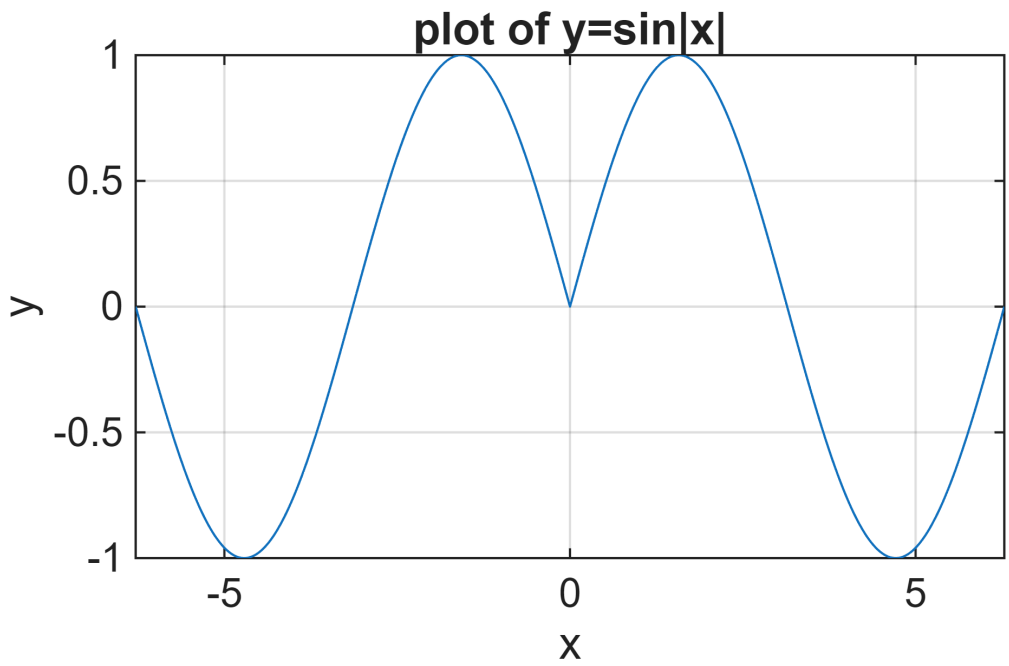
```
x=-2*pi:0.0001:2*pi
```

```
x = 1×125664
   -6.2832   -6.2831   -6.2830   -6.2829   -6.2828   -6.2827   -6.2826   -6.2825 ···
```

```
y=sin(abs(x))
```

```
y = 1×125664
   -0.0000   -0.0001   -0.0002   -0.0003   -0.0004   -0.0005   -0.0006   -0.0007 ···
```

```
plot(x,y)
xlabel('x')
ylabel('y')
title('plot of y=sin|x|')
grid on
```


plot of y=sin|x|
1
0.5
0
-0.5
-1
y
-5
0
5
x